DATA ANALYSIS REPORT

Decoding Leukemia from Gene Activity

A gene expression analysis of acute myeloid leukemia (AML) versus healthy blood and bone marrow cells, based on 64 samples and 22,283 gene probes (GEO GSE9476)

Prepared by

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1. Executive Summary

The problem. Acute myeloid leukemia (AML) is an aggressive blood cancer. Cancer cells differ from healthy cells in which genes are active, and measuring that activity could support diagnosis and a better understanding of the disease. We asked: which genes differ in AML, and is gene activity alone enough to tell AML apart from healthy blood and bone marrow cells?

What we analyzed. A public microarray dataset (GEO GSE9476) with 64 samples (26 AML and 38 healthy samples of four cell types) and 22,283 gene probes. The data was complete and consistent, so no samples or probes had to be removed. Two unusual blood samples were checked and kept because the difference appears biological.

Key results.

- **Strong molecular signal:** 163 probes differ clearly between AML and healthy cells (33 higher and 130 lower in AML), after correcting for multiple testing.
- **Highly accurate prediction:** two machine learning models identified the correct sample type for all 64 samples in stratified 5-fold cross-validation (100% accuracy).
- **AML is not one uniform group:** unsupervised clustering split AML samples across four groups, pointing to biological differences within AML.

Main recommendation. Treat these findings as a strong, internally validated starting point, not a clinical tool. The next step is to validate the model on independent datasets and to translate the significant probes into gene names and biological pathways.

2. Introduction and Scope

Every cell carries the same DNA, but different cells switch different genes on and off. Gene expression profiling measures how active thousands of genes are at once, giving a molecular fingerprint of a cell. In leukemia, this fingerprint can reveal what makes cancer cells different from the healthy blood-forming cells they come from.

Why this analysis. Knowing which genes behave differently in AML, and whether gene activity alone is enough to recognize it, is a first step toward better diagnostic markers and a better understanding of the disease. This analysis was carried out as part of a bioinformatics bootcamp project and answers four questions, labeled Q1 to Q4. They are answered directly in Section 5, and the recommendations in Section 6 are linked to them.

- Q1. Is the dataset clean and comparable enough for analysis?
- Q2. Do samples naturally group by cell type, based on gene expression alone?
- Q3. Which gene probes differ most between AML and healthy cells?
- Q4. Can a model predict the sample type from gene expression, and how reliable is it?

Scope. The analysis uses one public dataset (GSE9476) obtained from Kaggle, and covers data quality checks, exploratory analysis, dimensionality reduction, clustering, differential expression and classification. It does not include clinical outcomes, and gene probes were not mapped to gene names (see Limitations in Section 8).

3. Data Overview and Preparation

3.1 Dataset at a glance

Property	Description
Source	NCBI Gene Expression Omnibus, accession GSE9476, obtained via Kaggle
Samples	64 (one row per sample)
Gene probes	22,283 (Affymetrix microarray)
Sample types	AML (26), Bone_Marrow (10), PB, peripheral blood (10), PBSC_CD34 (10), Bone_Marrow_CD34 (8)
Values	Log-scale expression levels, from 3.19 to 14.47
Label column	"type", the cell type of each sample

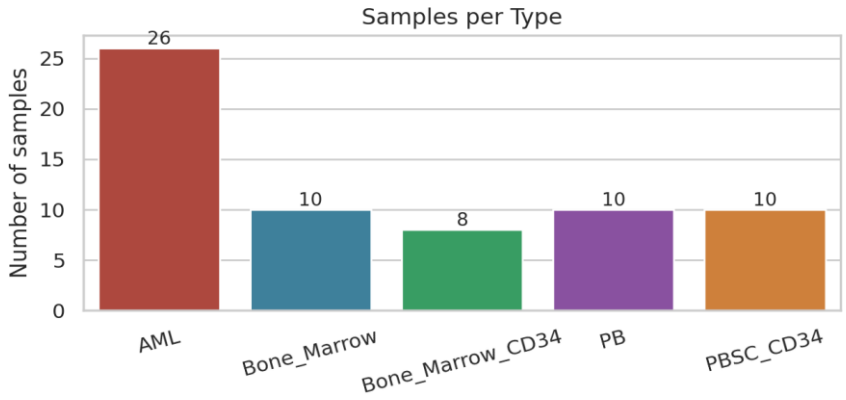


Figure 1. Number of samples per type. AML is the largest group; the healthy groups are small (8 to 10 samples each).

3.2 Data quality and cleaning

We checked the data for the usual problems before any analysis. The dataset was already clean, so no rows or columns were removed.

Check	Result	Action
Missing values	0	None needed
Duplicate rows	0	None needed
Constant probes (zero variance)	0	None needed
Negative or impossible values	0	None needed
Value range	3.19 to 14.47	Consistent with log-scale data
Unusual samples (outliers)	2 PB samples	Kept (see below)
Unusual values (outlier rule)	2.9% of values	Kept (see below)

Normalization check. For samples to be compared fairly, they must have a similar overall expression level. The median expression of every sample falls between 6.22 and 6.44 (Figure 2), so the samples are already normalized and technical differences between them are small. Unusual samples and values are examined next.

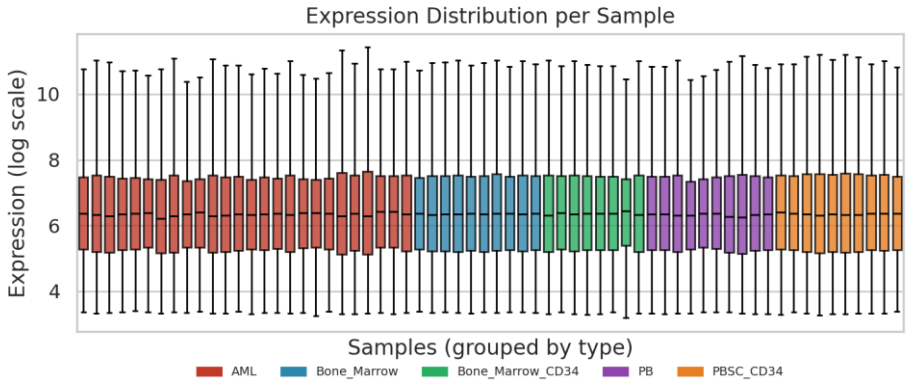


Figure 2. Distribution of expression values in each of the 64 samples, grouped by type. The boxes are almost aligned, which confirms comparable scales.

Outlier check. We looked for unusual samples and unusual values in two ways. First, for each sample we calculated its average similarity (correlation) to all other samples and converted it to a z-score, which shows how many standard deviations it lies from the group average; a score beyond -3 or +3 is a common warning sign. One peripheral blood (PB) sample scored -3.19, and a second PB sample was right at the limit (-3.00). Second, we counted individual expression values that fall outside the usual range of their probe (the standard 1.5 x interquartile range rule): they make up 2.9% of all values. We kept all 64 samples and all values. Both unusual samples are PB, the healthy type that is least similar to the others (Figure 5), so their difference may reflect real biology rather than a technical error, and removing them would shrink an already small group of 10 samples.

Preparation for modeling. For dimensionality reduction and clustering, each gene was standardized (mean 0, standard deviation 1) so that no gene dominates because of its scale. For classification, gene selection and scaling were performed inside each cross-validation fold, so the test samples never influenced which genes were chosen.

4. Statistical Analysis and Visual Exploration

4.1 Descriptive statistics

As an illustration, the first probe in the file (1007_s_at) has a mean expression of 7.82, a median of 7.81 and a standard deviation of 0.17. The mean and median are almost identical, which suggests a symmetric distribution, and the small spread shows that this probe changes little between samples. Probes like this carry little information; the analysis therefore focuses on probes that vary the most.

4.2 Correlation between variable genes

Computing correlations among all 22,283 probes would create a matrix of about 500 million values (roughly 4 GB), which is impractical. We therefore examined the 20 most variable probes. Their pairwise correlations range from -0.77 to 1.00, with an average absolute correlation of 0.54. Many genes rise and fall together, which means the data contains a smaller number of underlying patterns, and this makes dimensionality reduction effective.

4.3 Principal component analysis (PCA)

PCA summarizes thousands of genes into a few "components" that capture the largest differences between samples. The first component explains 19.5% of the variation, the second 14.7%, and the first ten together 65.7%. Reaching 80% requires 22 components, which shows that the variation is spread over many dimensions.

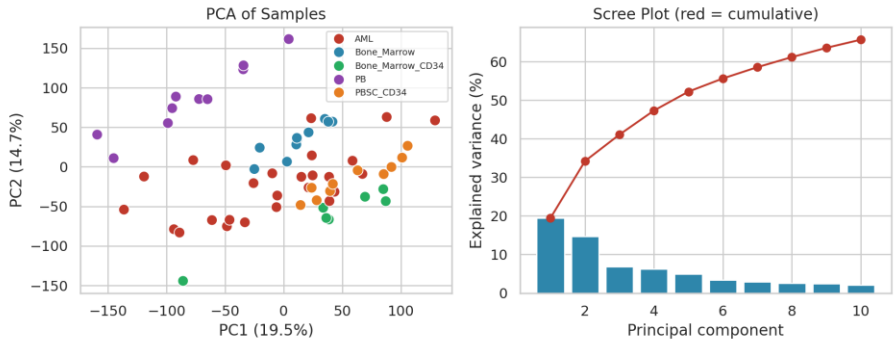


Figure 3. Left: samples plotted on the first two components, colored by true type. Right: variance explained by each component.

Even though PCA never saw the labels, structure is visible. Peripheral blood (PB) samples form a distinct group on the upper left, and the CD34 samples sit close together on the right. AML samples are spread over a wide area and overlap with several healthy types in these two dimensions, which already hints that AML is diverse.

4.4 Clustering with KMeans

To test the natural grouping further, KMeans (k = 5, using the first ten components) was run without labels and compared with the true types. The agreement is moderate (adjusted Rand index = 0.44; silhouette = 0.26).

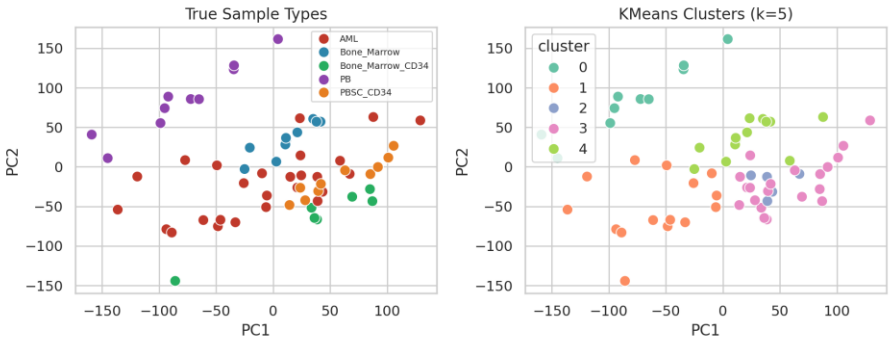


Figure 4. True sample types (left) compared with the five clusters found by KMeans (right).

True type	Cluster A	Cluster B	Cluster C	Cluster D	Cluster E
AML	0	14	5	4	3
Bone_Marrow	0	0	0	0	10
Bone_Marrow_CD34	0	1	0	7	0
PB	10	0	0	0	0
PBSC_CD34	0	0	0	10	0

Table 1. Number of samples of each true type in each KMeans cluster. Cluster letters are arbitrary labels.

All ten PB samples and all ten Bone_Marrow samples each fall into a single cluster, and the CD34 samples (PBSC_CD34 and seven of eight Bone_Marrow_CD34) are grouped together. AML, in contrast, is divided across four clusters (14, 5, 4 and 3 samples).

4.5 Similarity between sample types

Finally, we measured how similar samples are to each other in terms of gene expression, averaged by type (Figure 5). The diagonal shows similarity within a type; off-diagonal cells show similarity between types.

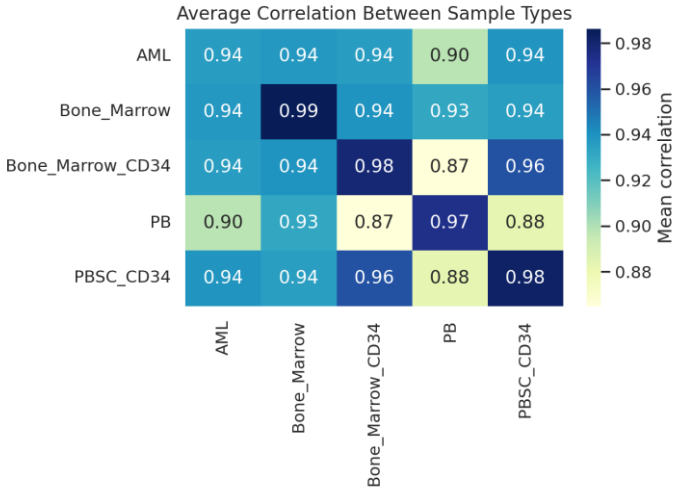


Figure 5. Average correlation of gene expression between and within sample types.

Healthy types are very uniform internally (0.97 to 0.99), whereas AML samples are only as similar to each other (0.94) as they are to healthy cells (0.94, except PB at 0.90). PB is the most distinct healthy type (0.87 to 0.93 with the others), and the two CD34 types are the closest pair (0.96).

5. Key Insights

Insight 1: AML has a clear but diverse molecular signature

Comparing 26 AML samples with the 38 healthy samples (Welch t-test with Benjamini-Hochberg correction, FDR < 0.05, and at least a two-fold change, i.e. |log2 fold change| ≥ 1) identified 163 probes: 33 with higher and 130 with lower expression in AML. A much larger set of 2,744 probes passes the significance test alone, which shows that leukemia changes gene activity widely, while a smaller set changes strongly.

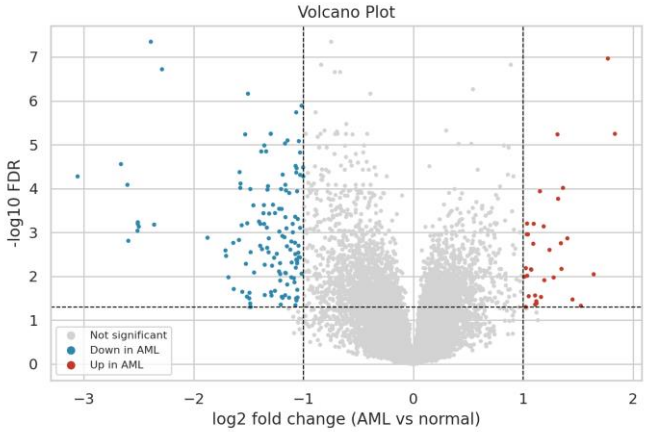


Figure 6. Volcano plot. Each dot is a probe; red = higher in AML, blue = lower in AML, grey = not significant by the thresholds.

Probe ID	log2 fold change	FDR	Direction	Rank
212224_at	-2.39	4.5E-08	Down in AML	1
206674_at	1.77	1.1E-07	Up in AML	2
213241_at	-2.29	1.9E-07	Down in AML	3
221477_s_at	-1.51	6.9E-07	Down in AML	4
203414_at	-1.02	1.3E-06	Down in AML	5
202459_s_at	-1.07	1.8E-06	Down in AML	6
218711_s_at	-1.30	5.6E-06	Down in AML	7
201362_at	-1.30	5.7E-06	Down in AML	8
201105_at	1.83	5.7E-06	Up in AML	9
210504_at	-1.53	5.8E-06	Down in AML	10

Table 2. The ten most significant probes. Probe IDs are Affymetrix identifiers, not gene names.

The strongest decrease is seen for probe 209116_x_at (log2 fold change -3.06, about an eight-fold reduction) and the strongest increase for probe 201105_at (+1.83, about a 3.6-fold increase). The most statistically significant probe is 212224_at.

Interpretation note: the healthy group combines several cell types, including immature CD34+ cells and mature blood cells. Some of the differences may therefore reflect cell maturity rather than leukemia itself, and should be confirmed with more specific comparisons.

Insight 2: Sample type can be predicted from gene expression

Two classifiers were trained to predict the five sample types from the 22,283 probes, using the 500 best probes selected inside each fold. Performance was measured with stratified 5-fold cross-validation, where every sample is predicted by a model that never saw it.

Model	Accuracy per fold	Mean accuracy
Logistic Regression	100% in all 5 folds	100%
Random Forest	100% in all 5 folds	100%

Table 3. Cross-validated classification performance.

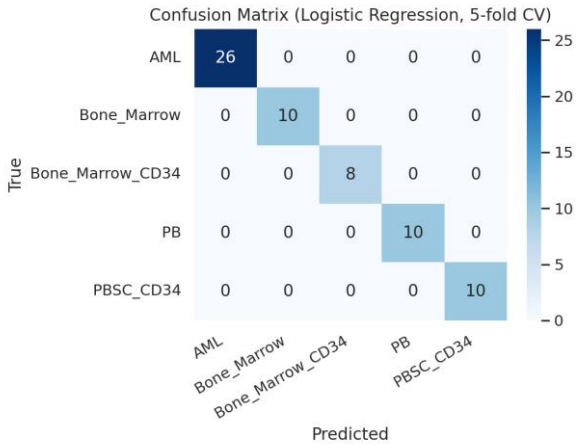


Figure 7. Confusion matrix of the Logistic Regression model. All 64 samples are on the diagonal, meaning every sample was classified correctly.

How to read this result. Perfect accuracy shows that the signal in this dataset is strong, but it does not prove that the model will work everywhere. The dataset is small (64 samples) and comes from a single study, so lab-specific effects may help the model. This is why external validation is the first recommendation below.

Insight 3: AML is heterogeneous

Three independent views agree. PCA shows AML samples spread widely, KMeans divides them across four groups, and the similarity matrix shows that AML samples are no more alike than AML and healthy cells. This suggests that AML contains molecular subgroups. Linking these subgroups to known AML subtypes or clinical outcomes would require clinical information that this dataset does not include.

Direct answers to the project questions

Project question	Direct answer	Where to see it
Q1. Is the dataset clean and comparable enough for analysis?	Yes. There are no missing, duplicate or constant values, and all samples have a similar overall expression level. Two unusual PB samples were checked and kept.	Section 3.2, Figure 2
Q2. Do samples group naturally by cell type?	Partly. PB and Bone_Marrow each form their own group, and the two CD34 types sit close together. AML does not form one group; it splits into four.	Sections 4.3 to 4.5, Table 1, Figure 5
Q3. Which probes differ most between AML and healthy cells?	163 probes: 33 higher and 130 lower in AML. Strongest decrease: 209116_x_at (-3.06). Strongest increase: 201105_at (+1.83). Most significant: 212224_at.	Insight 1, Figure 6, Table 2
Q4. Can a model predict the sample type, and how reliable is it?	Yes, within this dataset. Both models classified all 64 samples correctly (100% in 5-fold cross-validation), but this needs external validation before any real use.	Insight 2, Table 3, Figure 7

Table 4. Direct answers to the four project questions.

6. Actionable Recommendations

The following steps are ordered by priority. Each follows directly from the results and limits of this analysis, and the Why column shows which project question (Q1 to Q4) it relates to.

#	Recommendation	Why	Suggested next step
1	Validate the classifier on independent data	Q4: Results come from one study with 64 samples; perfect accuracy may not generalize.	Test on other public leukemia datasets from GEO
2	Map probes to gene names and pathways	Q3: Probe IDs cannot be interpreted biologically on their own.	Use the platform annotation, then run enrichment analysis on the 163 probes
3	Study AML subgroups	Q2: Clustering suggests AML is not uniform.	Obtain a larger AML cohort with clinical or subtype labels
4	Test a compact gene panel	Q3 and Q4: A small panel is cheaper and easier to use than 22,000 genes.	Test whether the top 20 to 50 probes keep the accuracy
5	Use more specific comparisons	Q3: Pooling healthy cell types may mix cell maturity with disease effects.	Compare AML with each healthy type separately
6	Check the two unusual blood samples	Q1: Two PB samples were less similar to the rest (z-scores -3.19 and -3.00).	Review the lab records, or repeat the main analysis with and without them.

Table 5. Recommended next steps.

7. Conclusion

Gene expression alone separated AML from healthy blood and bone marrow cells, revealed 163 strongly changed probes, and uncovered signs of hidden subgroups inside AML. With independent validation and gene-level annotation, this workflow can grow from a bootcamp analysis into a reliable foundation for leukemia research.

8. Appendix and Links

8.1 Links

- **Analysis notebook (Google Colab):** **https://colab.research.google.com/drive/1eMGJh13C1IUsyhyQlT5-GeJojQ6NvRCQ?usp=sharing**
- **Dataset (Kaggle):** https://www.kaggle.com/datasets/brunogrisci/leukemia-gene-expression-cumida
- **Original source:** NCBI Gene Expression Omnibus, accession GSE9476

8.2 Technical details

Item	Detail
Environment	Python in Google Colab (the full code is in the linked notebook)
Scaling	Each probe standardized so that no probe dominates because of its scale
PCA	First 10 components kept; a fixed random seed makes results repeatable
Outlier check	Average correlation of each sample with all others, as a z-score (warning sign beyond -3 or +3); 1.5 x interquartile range rule for individual values
Clustering	KMeans with 5 groups, applied to the first 10 components
Differential expression	Welch t-test per probe (AML vs all healthy samples), Benjamini-Hochberg FDR correction; thresholds: FDR below 0.05 and at least a two-fold change (log2 fold change of 1 or more)
Classification	Logistic Regression and Random Forest; the 500 best probes are selected inside each fold, and scaling is also done inside the fold for Logistic Regression
Validation	Stratified 5-fold cross-validation with shuffling and a fixed random seed

8.3 Limitations

- Small sample (64) from a single study, with no external validation set.
- Perfect cross-validated accuracy reflects a strong signal in this dataset and may not hold for other laboratories or platforms.
- Probes were not mapped to gene names, and no pathway analysis was done.

- AML was compared with all healthy types pooled together.
- The dataset has no clinical information, so AML subgroups could not be linked to outcomes.
- The two unusual PB samples were kept, and the analysis was not repeated without them.

8.4 Glossary

Term	Meaning
AML	Acute myeloid leukemia, an aggressive cancer of blood-forming cells
Gene probe	A measurement spot on a microarray that reports the activity of a gene
PCA	A method that summarizes many variables into a few components capturing the most variation
KMeans	A method that groups samples into a chosen number of clusters without using labels
FDR	False discovery rate, a correction that limits false positives when many genes are tested
log2 fold change	Difference in expression on a log2 scale; +1 means double, -1 means half
Cross-validation	Repeated training and testing on different parts of the data to estimate real performance
Outlier	A sample or value that lies unusually far from the rest of the data.
z-score	How many standard deviations a value lies from the average; beyond -3 or +3 is a common warning sign.